

Jaechul Roh

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EDUCATION

University of Massachusetts Amherst

Ph.D. Candidate in Computer Science

Advisor: Prof. [Amir Houmansadr](#)

GPA: 4.0/4.0

September 2023 – May 2027/8 (expected)

Amherst, Massachusetts, USA

University of Massachusetts Amherst

M.S. in Computer Science

Advisor: Prof. [Amir Houmansadr](#)

GPA: 4.0/4.0

September 2023 – May 2026

Amherst, Massachusetts, USA

Hong Kong University of Science and Technology

B.Eng. in Computer Engineering, School of Engineering

Final Year Thesis Advisor: Prof. [Jun Zhang](#)

*2 years of Compulsory Korean Military Duty

September 2017 – May 2023

Clear Water Bay, Hong Kong

RESEARCH INTERESTS

My research focuses on the **Safety & Security** of AI models and agents. Under the supervision of Prof. Amir Houmansadr, I study the trustworthiness of multimodal generative and interactive models across **audio**, **text**, and **vision**, with work published at NeurIPS, USENIX Security, and COLM. This work also includes ongoing collaborations with researchers at Google DeepMind and Qualcomm. Most recently, I completed a research internship at Dolby Laboratories, where I worked on the safety and security of real-time full-duplex speech models.

WORK EXPERIENCE

Dolby Laboratories

Ph.D. Research Intern, Supervisor: Deepak Chandran, Andrea Fanelli

- Worked on safety and security of real-time full-duplex speech models.

June 2026 – September 2026

San Francisco, California

Brave Software

Ph.D. Research Intern, Supervisor: Ali Shahin Shamsabadi

- Worked on web agent privacy.

June 2025 – September 2025

London, United Kingdom (Remote)

PUBLICATIONS

Selected Publication

- DuplexJail: Spoken Interruption Attacks on Full-Duplex Speech Models**
Jaechul Roh, Deepak Chandran, Amir Houmansadr, Andrea Fanelli
Preprint at arXiv
[\[paper\]](#) [\[demo\]](#)
- Benign Fine-Tuning Breaks Safety Alignment in Audio LLMs**
Jaechul Roh, Virat Shejwalkar, Amir Houmansadr
Preprint at arXiv
[\[paper\]](#)
- CodecAttack: Codec-Robust Latent-Space Prompt-Injection Attacks on Audio LLMs**
Jaechul Roh, Jean-Philippe Monteui, Jonathan Petit, Amir Houmansadr
Preprint at arXiv
[\[paper\]](#) [\[code\]](#)
- Multilingual and Multi-Accent Jailbreaking of Audio LLMs**
Jaechul Roh, Virat Shejwalkar, Amir Houmansadr
COLM 2025
[\[paper\]](#) [\[code\]](#)
- Backdooring Bias (B^2) into Stable Diffusion Models**
Ali Naseh, **Jaechul Roh**, Eugene Bagdasarian, Amir Houmansadr
USENIX Security '25

[\[paper\]](#) [\[code\]](#)

6. **OSLO: One-Shot Label-Only Membership Inference Attacks**
Yuefeng Peng, **Jaechul Roh**, Subhransu Maji, Amir Houmansadr
NeurIPS 2024
[\[paper\]](#)
7. **Bob's Confetti: Phonetic Memorization Attacks in Music and Video Generation**
Jaechul Roh, Zachary Novack, Yuefeng Peng, Niloofar Miresghallah, Taylor Berg-Kirkpatrick, Amir Houmansadr
Preprint at arXiv
[\[paper\]](#) [\[demo page\]](#) [\[video\]](#)
8. **OverThink: Slowdown Attacks on Reasoning LLMs**
Abhinav Kumar, **Jaechul Roh**, Ali Naseh, Marzna Karpinska, Mohit Iyyer, Amir Houmansadr, and Eugene Bagdasarian
Preprint at arXiv
[\[paper\]](#) [\[code\]](#)
9. **SPILLage: Agentic Oversharing on the Web**
Jaechul Roh, Eugene Bagdasarian, Hamed Haddadi, Ali Shain Shamsabadi
Preprint at arXiv
[\[paper\]](#) [\[code\]](#) [\[article\]](#)

INVITED TALKS

1. **AI Safety Night SF** **Aug 2026**
Invited Talk San Francisco
 - Provided a talk on modality-unique safety and security.
2. **Google Speech Technologies Group** **July 2025**
Invited Talk Google DeepMind
 - Presented our "Multilingual and Multi-Accent Jailbreaking of Audio LLMs" paper to the NFM Reading Group led by the Speech Technologies Group at Google DeepMind.

HONORS & AWARDS

2026 Apple Scholars in AI/ML PhD Fellowship **September 2025**
UMass Amherst Nominee

- Selected as departmental representative candidate for the fellowship

ACADEMIC SERVICES

Program Committee Member (Reviewer)

- **2026:** ICLR, ICASSP, ICML (**Best Reviewer**), COLM, NeurIPS
- **2025:** ICLR, COLM XLLM-Workshop

Article

- **Can You Trust What AI Hears (and Says)?**
Jaechul Roh
XRDS: Crossroads, The ACM Magazine for Students 31 (4), 34-39
[\[article\]](#)

SKILLS / LANGUAGES

Programming Language: Python, C++

Languages: Korean (Native), English (Native), Chinese (Fluent)